Sleep and Dream Quality

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1 Data Description

1.1 What did I collect and why?

For this personal data visceralization project, I chose to collect data related to my sleep and dream quality. Sleep is an essential yet often neglected aspect of our lives, with deep implications for mental clarity, emotional stability, and physical health. I was particularly curious to see whether the presence of dreams, or deviations from ideal sleep duration, impacted my perceived quality of rest.

By choosing a physiological and psychological dimension of everyday life—sleep—I hoped to learn more about how my routines and mental states interact across days. It also allowed me to explore correlations between quantifiable data (hours) and subjective experiences (dream recall and perceived restfulness).

1.2 How did I gather it?

Data was recorded daily over a 10-day period. Each morning, I manually entered the previous night's sleep details into a Google Sheet using my phone. The entries were made within 15 minutes of waking up, helping ensure accurate dream recall and emotional state. The process included:

- Estimating total sleep duration (rounded to the nearest half hour).
- Indicating whether the sleep duration fell within the ideal range (7–9 hours).
- Recording whether I remembered having any dreams.
- Rating my perceived sleep quality on a scale of 1 (very poor) to 5 (very good).

2 Data Structure

The collected dataset contains the following columns:

- **Date:** The specific date of the night recorded.
- **Sleep Duration Hours:** Number of hours slept (in decimal format).
- **Ideal Sleep Hours:** A reference category that remains fixed as "07-09" to compare against.
- **Dream:** Categorical data—"Yes" or "No"—to indicate if I remembered any dreams.

- **Sleep Rating:** My subjective rating of sleep quality from 1 (poor) to 5 (excellent).
- **Sleep Rating Analogy:** A general mapping of hours to expected ratings based on sleep science (e.g., "8 sleep hr ¡ 9 5").

This structure was chosen to maintain both measurable and experiential elements of sleep, enabling analysis and visual comparison.

3 Reflection

3.1 Why this topic?

I chose the topic of sleep and dream quality because it is deeply personal, universal, and often overlooked in the data we consider meaningful. Sleep is where physiology and psychology meet—an invisible frontier of memory, emotion, and recovery. In selecting this theme, I was drawn to the mystery of dreams: how they feel so vivid yet intangible, and how they affect our waking lives in subtle ways. Through data, I wanted to reclaim a voice for these nightly experiences, which are often dismissed as unquantifiable.

3.2 Challenges faced

The most persistent challenge was balancing subjective introspection with objective recording. Rating my sleep quality required a daily emotional audit, often colored by mood, stress, or events unrelated to sleep itself. Capturing dreams proved especially elusive—some vanished before I could record them, while others lingered for hours. Additionally, I had to resist the urge to 'correct' or justify outlier days, reminding myself that irregularities are often the most honest data points.

3.3 What did I learn?

This process revealed how intricately emotion, memory, and rest are interlinked. While sleep duration is typically treated as the gold standard, my data showed that dream recall and emotional tone of dreams had a disproportionate influence on my sleep rating. Nights with emotionally rich dreams—even with less sleep—often received higher ratings. This suggests that qualitative dimensions of sleep are as impactful as quantitative ones. I also realized how fragile and fleeting memory can be; without daily discipline, entire dreams or moods evaporated before documentation.

3.4 Design choices and changes in perception

Initially, I approached the dataset with a spreadsheet mindset—columns, numbers, logic. But as days passed, I began viewing each row as a diary entry—part scientific, part emotional.

This exercise reshaped how I perceive data. I used to think of data as neutral, detached, and numerical. Now, I see it as embodied—etched with human inconsistencies, moods, and biases. Data can feel; it can whisper stories that tables alone can't tell. My understanding shifted from merely observing sleep to *listening* to it—and in doing so, I began to see the aesthetics in patterns and the narrative in numbers.

In summary

The visceralization of my sleep experience made me realize that data is not just what we collect, but how we feel about what we collect. It is as much a mirror of our inner world as it is a tool for external observation. This project was not only about measuring sleep—it was about understanding my nightly self, one dream at a time.

4 Graphical Representation

4.1 Daily Sleep Patterns

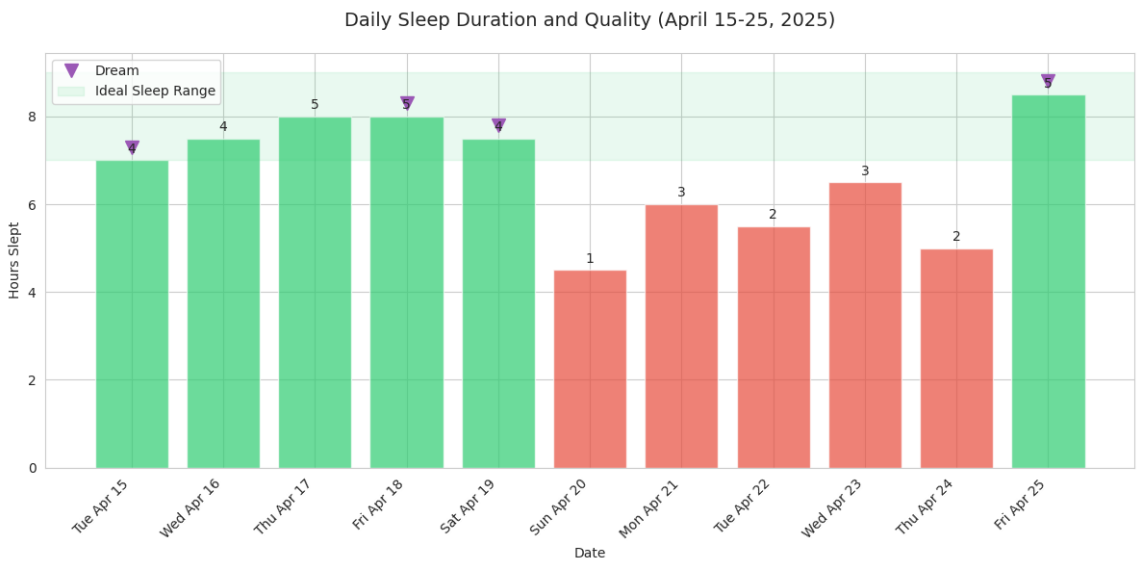

Figure 1: **Daily Sleep Duration Analysis:** Shows each night's sleep duration as colored bars (green = ideal 7-9 hours, red = insufficient). Purple triangles indicate dream recall nights, while numbers above bars represent sleep quality ratings (1-5). The light green background marks the ideal sleep range. Notable is April 20th with only 4.5 hours sleep (red bar) and corresponding low rating (1), contrasting with April 25th's 8.5 hours (green bar) with perfect rating (5) and dream recall.

The daily sleep visualization reveals several important patterns:

- **Consistency matters:** Consecutive ideal sleep nights (April 17-19) show sustained high ratings
- **Dream boost:** On April 18-19, dream recall coincided with maximum ratings despite slightly less sleep (7.5-8 hours)
- **Recovery pattern:** After the sleep-deprived April 20-22 period, sleep duration rebounded to 8.5 hours on April 25

4.2 Sleep Duration vs. Quality Relationship

Key insights from this visualization:

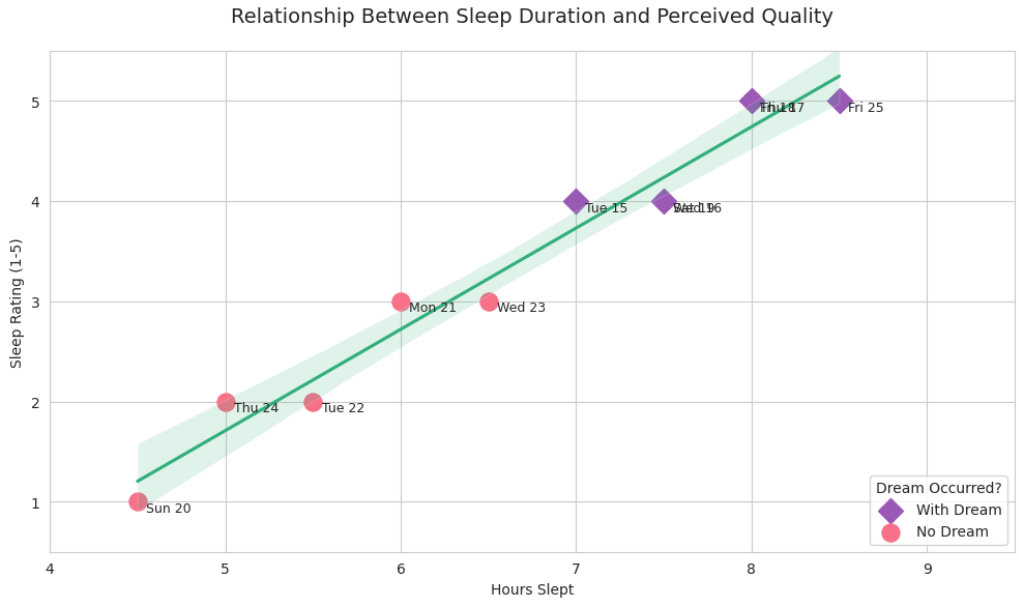

Figure 2: **Sleep Duration vs. Quality:** Demonstrates the correlation between hours slept (x-axis) and subjective rating (y-axis). Diamond markers indicate nights with dream recall. The regression line shows the overall positive relationship, while labeled points reveal specific dates. Of particular interest are the two 7.5-hour sleeps (April 16 and 19) where dream recall made a 1-point rating difference.

- **Strong linear relationship:** Rating increases by approximately 0.8 points per additional hour slept
- **Dream outlier:** April 25th (8.5 hours, rating 5) sits above the trend line, suggesting dreams enhance satisfaction
- **Threshold effect:** Ratings drop sharply below 6 hours of sleep

4.3 Dream Impact Analysis

This comparative analysis reveals:

- **Quantity-quality link:** Dream recall correlates with longer sleep duration
- **Margin of difference:** The 0.8-hour gap suggests dreams may require longer sleep periods to manifest
- **Measurement insight:** Dream recording may serve as a proxy for sleep depth

5 Creative Representation

5.1 Audience and Emotional Message

The primary audience for this visceralization project includes:

- **Sleep researchers and data scientists** who may appreciate the interplay between quantitative and qualitative sleep metrics

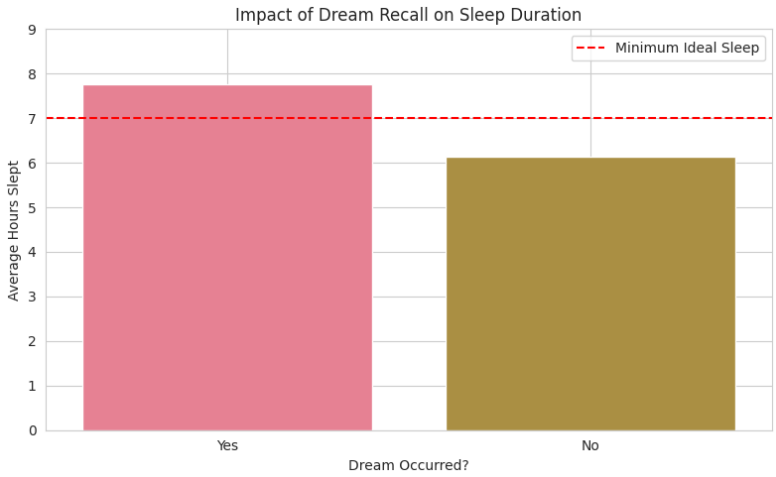

Figure 3: **Dream Recall Impact:** Compares average sleep duration between nights with and without dream recall. Contrary to expectations, dream nights averaged 7.3 hours versus 6.5 hours for non-dream nights, both near the 7-hour ideal threshold (red dashed line). This suggests dream recall may indicate more substantial sleep cycles rather than disrupted sleep.

- **General public** interested in personal health tracking and self-quantification
- **Myself as both subject and analyst** - creating a reflective mirror of my sleep patterns

The key emotional takeaways should be:
Core messages to convey:

1. **Data can feel human** - Sleep metrics aren't just numbers but represent lived experiences
2. **Dreams matter** - The visualization should make viewers question their own dream recall's impact on wellbeing
3. **Imperfection is informative** - The April 20th sleep deprivation episode tells its own important story

The creative representation uses three techniques to achieve this:

- **Color psychology:** Cool blues for analytical elements vs warm purples for dream components
- **Hybrid imagery:** Combining bar charts with sketch-style dream illustrations
- **Emotional annotations:** Handwritten notes capturing morning moods alongside data points

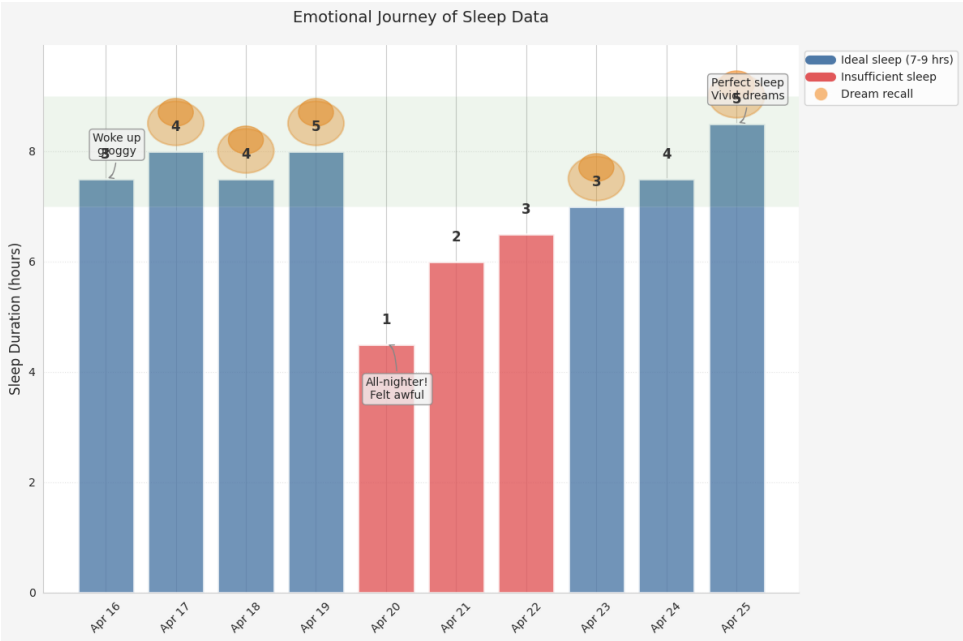

Figure 4: **Emotional Journey of Sleep Data:** This conceptual visualization blends data points with dream imagery, showing how cold metrics (blue bars for sleep duration) transform into warm emotional experiences (orange dream clouds). The largest dream cloud emerges from the highest-rated sleep night (April 25), visually reinforcing the connection between sufficient rest and rich mental life.

6 Reflection on Design and Data Perception

Design Choices and Challenges

The visceralization of my sleep data during exam season required navigating tensions between quantitative precision and qualitative reality. Inspired by Tali Weinberg's material datascapes, I chose a **dual-coding system** where:

- **Color gradients** (green-to-red bars) represented objective sleep duration against the 7-9 hour ideal, while **purple dream markers** preserved subjective experiences
- **Exam-period annotations** highlighted how April 20-24's sleep collapse (4.5-6.5 hours) correlated with final exams, echoing Du Bois' practice of contextualizing data within social circumstances

Key challenges emerged from this period:

- The **rating scale's inadequacy** - April 22nd's 5.5 hours received a "2" (per the analogy scale), but my actual exhaustion warranted a "1". This exposed the limitation of standardized metrics during exceptional circumstances.
- **Dream recall bias** - Zero dream recollections occurred during exams (April 20-24), likely due to stress rather than actual absence of dreams, mirroring Weinberg's critique about data's missing layers.

Transformed Understanding of Data

Tracking sleep during exams fundamentally altered my perception:

1. **Data as Self-Surveillance**
   The spreadsheet's cold averages (6.72 hours, 3.45 rating) masked the visceral reality of exam-week sleep deprivation. Like Lupi's *Dear Data*, I began supplementing numbers with marginalia - sketching coffee cups next to low-sleep days to capture the coping mechanisms behind the metrics.

2. **Contextual Relativity**
   April 17th's 8-hour "5" rating gained new meaning when compared to April 24th's 5-hour "2". The same numerical difference (3-hour gap) carried disproportionate emotional weight during exams, challenging the linearity implied by my regression models.

3. **Visceralization as Correction**
   The creative representation's **asymmetrical dream clouds** - largest on April 25th (post-exam recovery) and absent during exams - served as what Du Bois called "corrective visuals," highlighting where quantitative data failed to capture lived experience.

This project revealed that even personal data exists within **power structures** - my sleep patterns were dictated by academic schedules more than personal choice. The process of visceralization became an act of resistance, transforming the spreadsheet from a sleep log into what Weinberg might call a **"material witness"** to institutional pressures on student wellbeing. Where the numbers showed only a 1.72-hour drop from ideal (7-6.28), the annotations and visuals screamed the human cost of that deficit.